SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID:** SIH 26262
- **Problem Statement Title:** Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System
- **Add-on module:** SIH 26264, Integrated Institution Complaint Monitoring and Status Dashboard
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:** 199751
- **Team Name:** Phoenix Crew

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WAYMARK: WORKFLOW AUTOMATER

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PROBLEM

- Approvals cross many stages; follow-up runs on calls and spreadsheets.
- Nobody can say quickly which applications are stuck or which documents lapse next.
- Complaints on one institution reach different bureaus with no shared view.
- Nobody is clearly responsible, and the history is hard to piece together.

OUR APPROACH

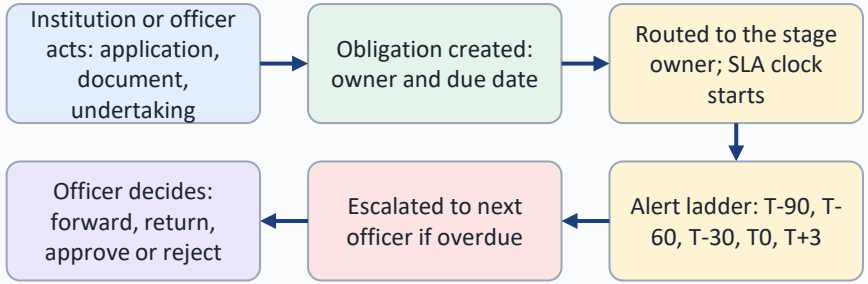
- Each deadline is an obligation with an owner, due date and alert ladder.
- Configurable stages and SLAs route applications and start the clock.
- A complaint dashboard shows which bureau holds each case.
- A flag says why it was raised. Officers make the call.

WHAT IS DIFFERENT

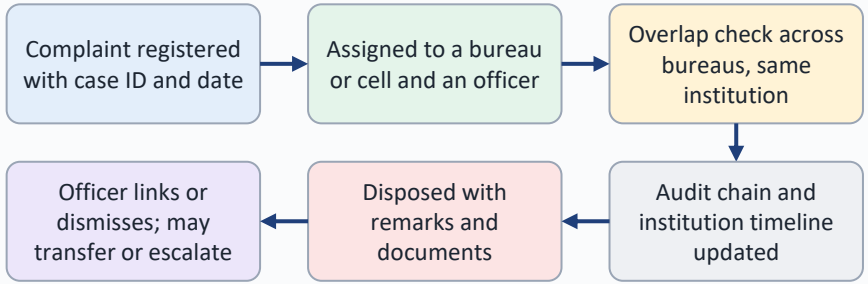
- One alert engine for approvals, documents, EVC visits and complaints
- Expiry dates are read from uploads and checked by staff
- Bureaus are warned about shared institutions even if the texts differ
- Audit log chained with SHA-256, so an edited row is caught

USER FLOW

Core approval flow (SIH 26262)



Add-on complaint flow (SIH 26264)



All changes are logged. Decisions stay with officers.

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TECHNICAL APPROACH

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INPUT

- Applications and stage actions
- Lease, fire NOC, FDR and other documents
- Undertakings and due dates
- EVC visit schedules
- Complaints (add-on)

Next.js forms, file upload

EXTRACTION

- OCR on scanned files
- Regex on text PDFs and notes
- Expiry date proposed with the matched snippet and a confidence
- Officer confirms or edits

pytesseract, pdf2image, regex

WORKFLOW ENGINE

- Stages, owner role and SLA days live in the database
- One Obligation entity for every deadline
- Alert ladder per obligation type
- Clock pauses on return for clarification

FastAPI, SQLAlchemy, APScheduler

MONITORING

- Pendency and ageing by stage and region
- Heuristic SLA-breach forecast
- Rule-based risk score with factors
- Complaint overlap (TF-IDF)

scikit-learn, plain rules

OUTPUT

- Role-based dashboards
- In-app and email alerts, SMS stub
- Reports with CSV export
- Institution timeline
- Audit trail

Recharts, SMTP, SHA-256 chain

ESCALATION LADDER

- Default steps (the admin can edit them):
- T-90, T-60, T-30: institution
- T0, T+3: regional officer
- T+10: admin. Missed steps are logged as skipped

DOCUMENT READING

- Upload > extract text > find dates
- Suggestion card shows the snippet
- Staff must confirm a date before it becomes an obligation

SECURITY

- JWT login, bcrypt passwords
- Role checks on every endpoint
- Complainant masking enforced in the API
- Hash-chained audit log

HUMANS DECIDE

- No automatic approval, rejection or penalty
- Each score shows the factors behind it
- Methods are named plainly: rules, TF-IDF, heuristic

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ADD-ON: SIH 26264 DASHBOARD

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Integrated Institution Complaint Monitoring and Status Dashboard: one window for complaints across RB, PGRC and other bureaus and cells.

Total received	Pending	Under process	Disposed	Transferred	Beyond timeline

EVERY CASE RECORD CARRIES

- Case ID, date of receipt, category, status
- Bureau or cell holding it now; assigned officer
- Last action, its date, and the next action required
- Expected disposal date and SLA countdown
- Transfer and escalation history, with reasons
- Remarks and supporting documents
- AICTE Permanent ID of the institution

Potential Overlapping Cases - Coordination Required

Institution A > PGRC > case under process

Institution A > RB > action initiated

Each bureau sees the other case's bureau, officer and status.

An officer links the cases or dismisses the alert. The system merges and closes nothing.

Institution-wise status (synthetic sample)

Institution	RB	PGRC	Other	Overlap
Kaveri Valley College	1	1	0	Alert
Northfield Institute	0	2	1	Alert
Malabar Pharmacy College	1	0	0	None

SEARCH, FILTERS AND REPORTS

- Search by institution name, Permanent ID or case ID
- Filter by bureau or cell, region, category, status, dates
- Reports by bureau, region and institution; ageing; disposal rate; overlaps; monthly and quarterly; CSV export

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OVERLAP, PRIVACY AND HISTORY

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Register: case ID, date, category → Assign bureau or cell and officer → Overlap check runs on save → Alert to every involved bureau → Officer links or dismisses → Transfer, escalate, dispose

TWO WAYS OVERLAP IS FOUND

1. Rule trigger
Open cases at two or more bureaus on one institution raise the alert. No text match needed.

2. Text similarity
TF-IDF cosine score from title and narrative, plus small bonuses for same category, recent date and a different bureau. Bonuses count only when the wording already overlaps. Suggested at 0.35 or above; the threshold is configurable.

Shared terms appear only after an officer confirms the link.

PRIVACY BY DESIGN

- Complainant name and contact: assigned bureau officer and admin only
- Regional officers see category, status, dates and a redacted summary
- Institution users see that a case exists, with category and status
- Masking is applied in the API serializer, not only the screen
- Reports leave out complainant fields
- Exports are logged

ACTION HISTORY AND STATUS

Each action stores officer, bureau, time, remarks and documents.

Transfer needs a reason. Escalation needs a written note.

Dashboard groups

- Pending: registered, assigned
- Under process: review, action initiated, escalated
- Transferred: moved, awaiting the new bureau
- Disposed: resolved, closed, rejected as frivolous

Beyond timeline: open and past the expected disposal date.

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FEASIBILITY AND VIABILITY

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Existing solution matrix

Factor	Spreadsheets and email	Existing portal	Waymark
Tracking	Per person, per file	Application status	Stage, owner and SLA clock
Deadline alerts	Manual follow-up	Manual follow-up	Ladders with escalation
Document expiry	Calendar notes	Submitted and checked apart	Read from file, then tracked
Complaints	One mailbox each	Existing channels	One cross-bureau view
Explanations	None	Not applicable	Each flag gives a reason
Licence cost	None	Not applicable	None, open source

WHY IT IS FEASIBLE

- Open-source stack: Next.js, FastAPI, SQL
- Rules and TF-IDF run offline without a GPU
- Built and tested one phase at a time
- Stages, SLAs and ladders can be edited

PRACTICAL IMPLEMENTATION

- Adapters ready for AICTE portal data
- Regional or state hosting; PostgreSQL
- Job queue and object storage at scale

Potential challenges	How we handle them
Scanned documents give wrong dates	Staff check every suggested date
Two cases wrongly flagged as duplicates	A flag is only a suggestion; an officer links the cases
Officers may distrust automated scores	Each score shows its factors and decisions stay with people
Handbook rules change	Stages, SLAs and ladders are admin-editable

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IMPACT AND BENEFITS

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IMPACTS

- Reminders reach the right person before a deadline passes
- Stuck stages and overdue items show up on one screen
- Bureaus find out about each other's cases before acting twice
- Open-source stack, so no licence fee
- Flags come with reasons, so staff can check them quickly

Feature	Impact	Benefit
Alert ladders	Reminders before, on and after the due date	Fewer missed deadlines
Stage SLA tracking	Late stages show days overdue and who owns them	Faster approvals
Document expiry reading	Date suggested from the file; staff confirm	Less manual data entry
EVC queue	Visits queued and tracked to report	Timely inspections
Overlap alert	Bureaus see shared-institution cases	No parallel or clashing action
Institution timeline	Approvals, documents and complaints together	One history per institution
Risk view	Weighted factors, all visible	Follow-up in a sensible order
Audit chain	Hash-chained log of every action	A record that shows if it was edited

Expected beneficiaries

AICTE officers
less chasing, clearer queues

EVC / SAC members
organised visit records

Bureau heads
cross-bureau complaint view

Institutions
clear status and deadlines

Administrators
reports and analytics

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RESEARCH AND REFERENCES

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1. **AI-Based Intelligent Document Management System (2023)**

Procedia Computer Science, Vol. 230, pp. 725-736. Relevance: document processing and information extraction.

2. **AI-Powered Intelligent Document Management System (2024)**

Information, Vol. 15, Issue 8, Article 461. Relevance: classifying and administering documents.

3. **Unveiling the Causes of Waiting Time in Business Processes (2024)**

Relevance: finding workflow delays and bottlenecks from event logs.

4. **Lightweight String Similarity Approaches for Duplicate Detection (2025)**

Journal of Informatics and Web Engineering, Vol. 4, Issue 3. Relevance: text similarity for duplicate complaints.

5. **PM4Py: A Process Mining Library for Python (2023)**

Software Impacts, Vol. 17, Article 100556. Relevance: process analysis and performance monitoring.

PROJECT LINKS

Source code
GitHub repository link (add before submitting)

Prototype images
Screenshots link (add before submitting)

Demo video
Walkthrough link (add before submitting)

OFFICIAL REFERENCE

AICTE Approval Process Handbook 2024-2027
Relevance: approval stages, inspections and institutional compliance. Stage names and SLAs in the prototype are illustrative defaults.

The prototype uses invented data and is not an official AICTE system.